

1 Problem 27: integer-valued polynomials over algebras

Let D be an integral domain with quotient field K . Following Problem 27 of Cahen–Fontana–Frisch–Glaz [1] (contributed by N. J. Werner), let A be a torsion-free D -algebra that contains D , is finitely generated as a D -module, and satisfies $K \cap A = D$; set $B = K \otimes_D A$. With a *central* indeterminate X and *right evaluation* $f(r) = \sum_i c_i r^i$ (coefficients on the left), one considers

$$\text{Int}(A) = \{ f \in B[X] : f(A) \subseteq A \}.$$

The set $\text{Int}(A)$ is always a left A -module, but when A is noncommutative it is not clear that it is closed under multiplication, hence not clear that it is a ring (see Werner [2] for the noncommutative integer-valued-polynomial framework).

Problem 27 has two parts: **(27a)** give an example of a D -algebra A with $\text{Int}(A)$ not a ring (possibly relaxing finite generation); and **(27b)**, where Werner [3] conjectured that $\text{Int}(A)$ is always a ring when D has finite residue rings, prove the conjecture or give a counterexample. We resolve both, with a single construction.

Before stating our proof, we establish the notations first. Let $S = \mathbb{F}_2[\varepsilon]/(\varepsilon^4)$. Inside $M_2(S)$, define

$$R = \left\{ \begin{pmatrix} \alpha + \eta\varepsilon^2 & \gamma\varepsilon + \lambda\varepsilon^3 \\ \delta\varepsilon + \mu\varepsilon^3 & \beta + \theta\varepsilon^2 \end{pmatrix} : \alpha, \beta, \gamma, \delta, \eta, \theta, \lambda, \mu \in \mathbb{F}_2 \right\}. \quad (1.1)$$

We use the following elements of R :

$$\begin{aligned} e &= \begin{pmatrix} 1 & 0 \\ 0 & 0 \end{pmatrix}, & f &= \begin{pmatrix} 0 & 0 \\ 0 & 1 \end{pmatrix}, \\ u &= \begin{pmatrix} 0 & \varepsilon \\ 0 & 0 \end{pmatrix}, & v &= \begin{pmatrix} 0 & 0 \\ \varepsilon & 0 \end{pmatrix}. \end{aligned}$$

Define

$$p = uv, \quad q = vu, \quad s = uvu, \quad w = vuw.$$

Explicitly,

$$p = \begin{pmatrix} \varepsilon^2 & 0 \\ 0 & 0 \end{pmatrix}, \quad q = \begin{pmatrix} 0 & 0 \\ 0 & \varepsilon^2 \end{pmatrix},$$

and

$$s = \begin{pmatrix} 0 & \varepsilon^3 \\ 0 & 0 \end{pmatrix}, \quad w = \begin{pmatrix} 0 & 0 \\ \varepsilon^3 & 0 \end{pmatrix}.$$

Now let

$$D = \mathbb{F}_2[\pi], \quad K = \mathbb{F}_2(\pi),$$

and define

$$A = D \otimes_k R, \quad B = K \otimes_D A.$$

Note that the symbols π and ε have different meanings: π is the indeterminate in the domain D , while ε is nilpotent in S . In the polynomial ring $B[X]$, the indeterminate X commutes with all coefficients. Define

$$\Phi(X) = uX^2 + eX^3 + (e + u)X^4 + eX^5 + eX^6, \quad (1.2)$$

and set

$$P(X) = \frac{\Phi(X)}{\pi}, \quad Q(X) = e. \quad (1.3)$$

We will prove

$$P, Q \in \text{Int}(A) \quad \text{but} \quad PQ \notin \text{Int}(A).$$

More precisely, the proof will establish

$$\Phi(r) = 0 \quad \text{for every } r \in R,$$

while

$$(\Phi(X)e)(u+v) = s \neq 0.$$

The first identity implies that Φ/π is integer-valued on A . The second identity implies that $(\Phi/\pi)e$ is not integer-valued.

Theorem 1.1. *The domain $D = \mathbb{F}_2[\pi]$ has finite residue rings, and A is a torsion-free noncommutative D -algebra, free of rank 8, satisfying*

$$K \cap A = D$$

inside $B = K \otimes_D A$. Moreover,

$$P, Q \in \text{Int}(A), \quad PQ \notin \text{Int}(A).$$

Consequently, $\text{Int}(A)$ is not a ring. Thus A gives the example requested in Problem 27a and disproves the conjecture in Problem 27b.

Proof. We first verify that R is a finite noncommutative algebra.

Lemma 1.2. *The set R is a unital noncommutative $\mathbb{F}_2$ -algebra of dimension 8. A $\mathbb{F}_2$ -basis is*

$$e, f, u, v, p, q, s, w.$$

Consequently,

$$|R| = 2^8 = 256.$$

Proof. Write

$$S_{\text{ev}} = \mathbb{F}_2 + \mathbb{F}_2\varepsilon^2, \quad S_{\text{odd}} = \mathbb{F}_2\varepsilon + \mathbb{F}_2\varepsilon^3.$$

Then R consists of matrices whose diagonal entries belong to S_{ev} and whose off-diagonal entries belong to S_{odd} . We have

$$S_{\text{ev}}S_{\text{ev}} \subseteq S_{\text{ev}}, \quad S_{\text{odd}}S_{\text{odd}} \subseteq S_{\text{ev}},$$

and

$$S_{\text{ev}}S_{\text{odd}} \subseteq S_{\text{odd}}, \quad S_{\text{odd}}S_{\text{ev}} \subseteq S_{\text{odd}}.$$

It follows from the usual formula for matrix multiplication that R is closed under multiplication. It is clearly closed under addition, and it contains

$$e + f = \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}.$$

Thus R is a unital subalgebra of $M_2(S)$. The eight elements

$$e, f, u, v, p, q, s, w$$

correspond to the eight independent coefficients in (1.1), so they form a $\mathbb{F}_2$ -basis. Finally,

$$uv = p = \begin{pmatrix} \varepsilon^2 & 0 \\ 0 & 0 \end{pmatrix},$$

whereas

$$vu = q = \begin{pmatrix} 0 & 0 \\ 0 & \varepsilon^2 \end{pmatrix}.$$

Therefore $uv \neq vu$, so R is noncommutative. □

We next prove the crucial fact that Φ vanishes on every element of R .

Lemma 1.3. *For every $r \in R$,*

$$\Phi(r) = 0.$$

Proof. Take an arbitrary element

$$r = \begin{pmatrix} \alpha + \eta\varepsilon^2 & \gamma\varepsilon + \lambda\varepsilon^3 \\ \delta\varepsilon + \mu\varepsilon^3 & \beta + \theta\varepsilon^2 \end{pmatrix} \in R.$$

Set

$$\rho = 1 + \alpha + \beta, \quad z = \eta + \theta.$$

We shall calculate

$$r^2 + r^4$$

and

$$r^3 + r^4 + r^5 + r^6.$$

Because S is commutative, the Cayley–Hamilton identity applies to the matrix r . Let

$$\tau = \text{trace}(r) = \alpha + \beta + z\varepsilon^2$$

and

$$\Delta = \det(r) = \alpha\beta + (\alpha\theta + \beta\eta + \gamma\delta)\varepsilon^2.$$

Since the characteristic is 2, the Cayley–Hamilton identity becomes

$$r^2 = \tau r + \Delta I_2.$$

We also have

$$\tau^2 = \alpha + \beta, \quad \Delta^2 = \alpha\beta,$$

because $\varepsilon^4 = 0$. Squaring the identity

$$r^2 = \tau r + \Delta I_2$$

and using $2 = 0$ in $\mathbb{F}_2$, we obtain

$$r^4 = \tau^2 r^2 + \Delta^2 I_2 = \tau^3 r + (\tau^2 \Delta + \Delta^2) I_2.$$

It follows that

$$r^2 + r^4 = \tau(1 + \tau^2)r + (\Delta(1 + \tau^2) + \Delta^2)I_2.$$

Since

$$1 + \tau^2 = 1 + \alpha + \beta = \rho,$$

this calculation simplifies to

$$r^2 + r^4 = \rho \begin{pmatrix} \gamma\delta\varepsilon^2 & \gamma z\varepsilon^3 \\ \delta z\varepsilon^3 & \gamma\delta\varepsilon^2 \end{pmatrix}. \quad (1.4)$$

To see the simplification holds, if $\rho = 0$, then $\alpha \neq \beta$, and hence

$$\alpha\beta = 0.$$

The preceding Cayley–Hamilton expression is therefore zero. If $\rho = 1$, then $\alpha = \beta$. Write $\alpha = \beta = a$. Then $\tau = z\varepsilon^2$, and

$$\Delta = a + (az + \gamma\delta)\varepsilon^2.$$

Substitution into the preceding expression gives exactly (1.4).

Now observe that

$$r^3 + r^4 + r^5 + r^6 = (r + r^2)(r^2 + r^4).$$

If $\rho = 0$, then (1.4) shows that this expression is zero. Suppose that $\rho = 1$. Then $\alpha = \beta$. Every entry of $r^2 + r^4$ is divisible by ε^2 . Therefore, when computing

$$(r + r^2)(r^2 + r^4),$$

only the terms of degree at most 1 in $r + r^2$ can contribute, because every term containing ε^4 is zero. Modulo $\varepsilon^2 M_2(S)$, we have

$$r + r^2 \equiv \begin{pmatrix} 0 & \gamma\varepsilon \\ \delta\varepsilon & 0 \end{pmatrix}.$$

Multiplying this matrix by the matrix in (1.4) gives

$$r^3 + r^4 + r^5 + r^6 = \rho \begin{pmatrix} 0 & \gamma\delta\varepsilon^3 \\ \gamma\delta\varepsilon^3 & 0 \end{pmatrix}. \quad (1.5)$$

We now return to the definition of Φ . Regrouping its terms gives

$$\begin{aligned} \Phi(r) &= ur^2 + er^3 + (e + u)r^4 + er^5 + er^6 \\ &= u(r^2 + r^4) + e(r^3 + r^4 + r^5 + r^6). \end{aligned}$$

Using (1.4), we obtain

$$u(r^2 + r^4) = \rho\gamma\delta \begin{pmatrix} 0 & \varepsilon^3 \\ 0 & 0 \end{pmatrix} = \rho\gamma\delta s.$$

Using (1.5), we obtain

$$e(r^3 + r^4 + r^5 + r^6) = \rho\gamma\delta \begin{pmatrix} 0 & \varepsilon^3 \\ 0 & 0 \end{pmatrix} = \rho\gamma\delta s.$$

Therefore

$$\Phi(r) = \rho\gamma\delta s + \rho\gamma\delta s.$$

Since the characteristic is 2,

$$x + x = 0$$

for every $x \in R$. Hence

$$\Phi(r) = 0.$$

Since r was arbitrary, Φ vanishes on all of R . □

Now we show that, although Φ vanishes everywhere, multiplying it on the right by the constant polynomial e destroys this property.

Lemma 1.4. $(\Phi(X)e)(u + v) = s \neq 0$.

Proof. Since X commutes with the coefficients,

$$\begin{aligned}\Phi(X)e &= (ue)X^2 + (e^2)X^3 + ((e + u)e)X^4 \\ &\quad + (e^2)X^5 + (e^2)X^6.\end{aligned}$$

The relevant multiplication rules are

$$ue = 0, \quad e^2 = e, \quad (e + u)e = e.$$

Therefore

$$\Phi(X)e = eX^3 + eX^4 + eX^5 + eX^6. \tag{1.6}$$

Now let

$$a_0 = u + v.$$

Because $u^2 = v^2 = 0$, we have

$$a_0^2 = (u + v)^2 = uv + vu = p + q.$$

Consequently,

$$\begin{aligned}a_0^3 &= (p + q)(u + v) \\ &= pu + pv + qu + qv \\ &= s + w.\end{aligned}$$

Indeed,

$$pu = s, \quad qv = w, \quad pv = qu = 0.$$

Also we know $a_0^4 = 0$ because every term in a_0^4 contains ε^4 . Thus

$$a_0^5 = a_0^6 = 0.$$

Using (1.6), we get

$$\begin{aligned}(\Phi(X)e)(a_0) &= ea_0^3 + ea_0^4 + ea_0^5 + ea_0^6 \\ &= e(s + w).\end{aligned}$$

Since

$$es = s, \quad ew = 0,$$

we conclude that

$$(\Phi(X)e)(a_0) = s.$$

Finally,

$$s = \begin{pmatrix} 0 & \varepsilon^3 \\ 0 & 0 \end{pmatrix} \neq 0,$$

because $\varepsilon^3 \neq 0$ in

$$S = \mathbb{F}_2[\varepsilon]/(\varepsilon^4).$$

□

We are now ready to prove the full Theorem.

Proof of Theorem 1.1. By Lemma 1.2,

$$A = De \oplus Df \oplus Du \oplus Dv \oplus Dp \oplus Dq \oplus Ds \oplus Dw.$$

Therefore A is a free D -module of rank 8. In particular, A is finitely generated and torsion-free over D . The scalar embedding is

$$D \longrightarrow A, \quad d \longmapsto d(e + f).$$

Since $e + f$ is the identity of R , the image of D lies in the center of A . We also have

$$B = K \otimes_D A \cong K \otimes_k R.$$

Suppose that a scalar $\kappa \in K$ belongs to A . As an element of B , this scalar is

$$\kappa(e + f) = \kappa e + \kappa f.$$

In the displayed D -basis of A , the coefficients of both e and f are κ . Since coefficients of elements of A must lie in D , we obtain $\kappa \in D$. Thus we have $K \cap A = D$.

Next, $D = \mathbb{F}_2[\pi]$ has finite residue rings. Indeed, let

$$0 \neq d(\pi) \in D$$

have degree m . Every residue class in D/dD has a representative of degree less than m . There are exactly 2^m such polynomials. Therefore D/dD is finite. If d is a unit, the quotient is the one-element zero ring.

It remains to prove the statements concerning P and Q . Reduction modulo π gives

$$A/\pi A \cong (D/\pi D) \otimes_k R \cong k \otimes_k R \cong R.$$

For $a \in A$, denote its image in R by $\bar{a}$. Reduction modulo π commutes with addition, multiplication, powers, and polynomial evaluation. Therefore

$$\overline{\Phi(a)} = \Phi(\bar{a}).$$

By Lemma 1.3,

$$\Phi(\bar{a}) = 0.$$

Hence

$$\overline{\Phi(a)} = 0.$$

The kernel of the reduction map

$$A \longrightarrow A/\pi A$$

is πA . Thus

$$\Phi(a) \in \pi A.$$

Consequently,

$$P(a) = \frac{\Phi(a)}{\pi} \in A.$$

This holds for every $a \in A$, so $P \in \text{Int}(A)$. The polynomial $Q = e$ is constant and has value in A . Therefore $Q \in \text{Int}(A)$. Finally, evaluate the product at

$$a_0 = u + v.$$

Since

$$P(X)Q(X) = \frac{\Phi(X)e}{\pi},$$

Lemma 1.4 gives

$$(PQ)(a_0) = \frac{(\Phi(X)e)(a_0)}{\pi} = \frac{s}{\pi}.$$

We claim that

$$\frac{s}{\pi} \notin A.$$

Every element of A has a unique expression

$$d_1e + d_2f + d_3u + d_4v + d_5p + d_6q + d_7s + d_8w,$$

with $d_1, \dots, d_8 \in D$. The coefficient of s in s/π is $1/\pi$. But

$$\frac{1}{\pi} \notin \mathbb{F}_2[\pi].$$

Indeed, if

$$\frac{1}{\pi} = g(\pi)$$

for some $g(\pi) \in \mathbb{F}_2[\pi]$, then $\pi g(\pi) = 1$. The left-hand side has constant term 0, while the right-hand side has constant term 1, which is impossible. Therefore

$$\frac{s}{\pi} \notin A.$$

It follows that $(PQ)(a_0) \notin A$, and hence $PQ \notin \text{Int}(A)$. Thus $\text{Int}(A)$ is not closed under multiplication. □

□

References

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